

Control Systems

1. Control System – System that manages, commands, or regulates behavior.
2. Open Loop System – No feedback; output doesn't affect input.
3. Closed Loop System – Uses feedback to control output.
4. Feedback – Portion of output returned to input.
5. Transfer Function – Ratio of output to input (Laplace domain).
6. Block Diagram – Graphical representation of system components.
7. Signal Flow Graph – Representation of signal paths in a system.
8. Gain – Ratio of output to input amplitude.
9. Stability – Ability of system to return to equilibrium.
10. Time Response – Output behavior over time.
11. Transient Response – Short-term response before steady state.
12. Steady State Response – Long-term response after settling.
13. Rise Time – Time to reach from 10% to 90% of final value.
14. Settling Time – Time to stay within acceptable error band.
15. Peak Time – Time to reach maximum overshoot.
16. Overshoot – Exceeding final value in response.
17. Damping Ratio (ζ) – Measures oscillations in system.
18. Natural Frequency (ω_n) – Frequency of system oscillation.
19. Pole – Value making denominator zero.
20. Zero – Value making numerator zero.
21. Root Locus – Path of poles with varying gain.
22. Bode Plot – Frequency response (magnitude & phase plot).
23. Nyquist Plot – Stability analysis in frequency domain.
24. PID Controller – Proportional-Integral-Derivative controller.
25. Proportional Control – Output proportional to error.
26. Integral Control – Eliminates steady-state error.
27. Derivative Control – Predicts future error trend.
28. Laplace Transform – Converts time domain to s-domain.
29. State Space – Representation using state variables.
30. Controllability – Ability to control system states.